F79SP Session 5: Ruin Problem on Simple Random Walks

Recall previously for general quantum walk with $a > 0$ and $\mu = \mathbf{E}\xi_\mathbf{i}$, we have:

- if $\mu < 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\,T_R < \infty$;
- if $\mu = 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\,T_R = \infty$;
- if $\mu > 0$, then $\mathbf{P}(R) < 1$ and $\mathbf{E}\,T_R = \infty$.

Our goal is to obtain exact formula of the highlighted results, for simple random walks.

1 Ruin probability when $\mu = p - q > 0$

Main result: For SRW $\{S_n\}_{n\geq 0}$ with $S_0 = a > 0$ and $\mathbf{P}(\xi_i = 1) = p$ and $\mathbf{P}(\xi_i = -1) = q$:

$$\mathbf{P}(R) = \left(\frac{q}{p}\right)^a \text{ for } p > \frac{1}{2}.$$

Idea of proof:

- We consider the events $E_{b,a} := \{\text{process } \{S_n\} \text{ hits state 0 before state } b \,|\, S_0 = a\}$ and $\alpha_{b,a} = \mathbf{P}(E_{b,a})$.
- $E_{b,a}$ are increasing in b and thus their probability $\alpha_{b,a}$ too.
- We can think of the event, ruin, as $R = \lim_{b\to\infty} E_{b,a}$, and thus $\mathbf{P}(R) = \lim_{b\to\infty} \alpha_{b,a}$.
- It is then left to us that what is $\alpha_{b,a}$?

The Gambler's ruin problem

Consider a gambling game:

- You have an initial capital of $a \in \mathbb{Z}^+$.
- On any turn you win \$1 with probability p or lose \$1 with probability $q = 1 - p$.
- You will quit playing if your fortune reaches b.
- If your fortune reaches 0, then the casino makes you stop.
- We called 0 and b as absorbing barriers of the process.

Note that there would be a total of three possible outcomes below:

- Random walk hits state 0 before state b.
- Random walk hits state b before state 0.
- Random walk stay in between 0 and b forever.

The notation $E_{b,a}$ can be think of as the event of you lost all your money to the casino. We are interested with $\alpha_{b,a} = \mathbf{P}(E_{b,a})$.

Theorem 1.1. *Consider a simple random walk with $a > 0$, and $b > a$. Then:*

$$\alpha_{b,a} = \mathbf{P}(\textit{process } \{S_n\} \textit{ hits state } 0 \textit{ before state } b \,|\, S_0 = a) = \begin{cases} \frac{(q/p)^a - (q/p)^b}{1-(q/p)^b} & p \neq \frac{1}{2} \\ 1 - \frac{a}{b} & p = \frac{1}{2}. \end{cases}$$

Proof. Before proving, note that some obvious info of $\alpha_{b,a}$ we know are: $\alpha_{b,0} = 1$ and $\alpha_{b,b} = 0$.
Outline of proof: (See full proof in notes)

1. We use **conditioning on first step technique** to obtain $\alpha_{b,a} = p\alpha_{b,a+1} + q\alpha_{b,a-1}$ (for any a between 0 and b)

$$\begin{aligned} \alpha_{b,a} &= \mathbf{P}(\{S_n\} \text{ hits 0 before } b \,|\, S_0 = a) \\ &= \mathbf{P}(S_1 = a+1 | S_0 = a)\mathbf{P}(\{S_n\} \text{ hits 0 before } b \,|\, S_1 = a+1, S_0 = a) \\ &+ \mathbf{P}(S_1 = a-1 | S_0 = a)\mathbf{P}(\{S_n\} \text{ hits 0 before } b \,|\, S_1 = a-1, S_0 = a) \end{aligned}$$

2. The equation then **iteratively** gives the following: $\alpha_{b,a} - \alpha_{b,a+1} = \left(\frac{q}{p}\right)^a (\alpha_{b,0} - \alpha_{b,1})$ (for any a between 0 and b)

3. We use **method of difference** and 2. to obtain:

$$\begin{aligned} \alpha_{b,0} - \alpha_{b,a} &= (\alpha_{b,a-1} - \alpha_{b,a}) + (\alpha_{b,a-2} - \alpha_{b,a-1}) + (\alpha_{b,a-3} - \alpha_{b,a-2}) + \cdots + (\alpha_{b,0} - \alpha_{b,1}) \\ &= \left(\left(\frac{q}{p}\right)^{a-1} + \left(\frac{q}{p}\right)^{a-2} + \left(\frac{q}{p}\right)^{a-3} + \cdots + \left(\frac{q}{p}\right)^{0} \right) (\alpha_{b,0} - \alpha_{b,1}) \end{aligned}$$

4. Referring to 3., we wish to compute $\alpha_{b,a}$ and we already knew that $\alpha_{b,0} = 1$. We are left with the value of $\alpha_{b,1}$. This can be computed by substituting $a = b$ into the equation in 3., which gives $\alpha_{b,a} = \alpha_{b,b} = 0$. Once obtain $\alpha_{b,1}$, sub into 3., we obtain the formula of $\alpha_{b,a}$.

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Conclusion: $\mathbf{P}(R) = \lim_{b\to\infty} \alpha_{b,a} = \begin{cases} \left(\frac{q}{p}\right)^a & p > \frac{1}{2} \\ 1 & p \leq \frac{1}{2}. \end{cases}$. We can express:

Go through Example in notes, pg 3.9.

Further discussion

We can find $\beta_{b,a} = \mathbf{P}(\text{process } \{S_n\} \text{ hits state } b \text{ before state } 0 \,|\, S_0 = a)$ using **symmetry technique**.

$\beta_{b,a}$ can be obtained by replacing formula of $\alpha_{b,a}$ with:

- $q \to p$
- $p \to q$
- $a \to b - a$

The final form of $\beta_{b,a}$ is:

$$\beta_{b,a} = \begin{cases} \frac{(p/q)^{b-a} - (p/q)^b}{1-(p/q)^b} & p \neq \frac{1}{2} \\ 1 - \frac{b-a}{b} & p = \frac{1}{2}. \end{cases}$$

. It can be further verified that:

$$\beta_{b,a} + \alpha_{b,a} = 1.$$

This means that the random walk must hit atleast one of the barriers. The probability of random walk stay in between 0 and b forever is 0.

2 Expected time to ruin ($p < \frac{1}{2}$)

Main result: For SRW $\{S_n\}_{n\geq 0}$ with $S_0 = a > 0$ and $\mathbf{P}(\xi_i = 1) = p$ and $\mathbf{P}(\xi_i = -1) = q$:

$$\mathbf{E}(T_R) = \frac{a}{q-p}\,, \text{ if } p < \frac{1}{2}.$$

Proof. Define some notations:

- Let time to ruin $T_R = T_a$ when $S_0 = a$.
- Let $T_{i,i-1}$ be the first time to get to level $i-1$ for a random walk that starts from i.

Outline of proof:

1. Observe that $T_a = T_{a,a-1} + T_{a-1,a-2} + \cdots + T_{1,0}$ (By time homogeneous property)
2. By space homogeneous property and independence of increments, we know each interval $T_{i,i-1}$ involved are of i.i.d.. Therefore, $\mathbf{E}T_{i,i-1} = \mathbf{E}T_{1,0}$.
3. Together, 1. and 2. give $\mathbf{E}T_a = a\mathbf{E}T_{1,0}$, hence we are left to determine $\mathbf{E}T_{1,0} = \mathbf{E}T_1$.
4. To determine $\mathbf{E}T_{1,0}$, we use **conditioning on first step technique**:

- Define $\tilde{T}_k = \{\min n \geq 0 \text{ s.t. } S_{n+1} = 0 \,|\, S_1 = k\}$.
- We have $T_1 = 1 + \mathbf{I}(\xi_1 = 1)\tilde{T}_2 + \mathbf{I}(\xi_1 = -1)\tilde{T}_0$.
  Comment: The time to ruin from $S_0 = 1$ is $1 + \tilde{T}_2$ if $\xi_1 = 1$, otherwise, $1 + \tilde{T}_0$ if $\xi_1 = -1$. Use indicator random variable $\mathbf{I}$ for partitioning cases.
- Taking expectation gives:

5. This gives $\mathbf{E}T_1 = \frac{1}{1-2p}$ and thus $\mathbf{E}T_a = \frac{a}{1-2p} = \frac{a}{q-p}$ by 3..

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Post note: For an event A, the indicator random variable for A is defined as:

$$\mathbf{I}(A) = \begin{cases} 1, & \text{if event } A \text{ happens,} \\ 0, & \text{if event } A \text{ does not happen.} \end{cases}$$

Note that: $\mathbf{E}(\mathbf{I}(A)) = 1\mathbf{P}(A) + 0\mathbf{P}(A^c) = \mathbf{P}(A)$.